



Characterization of the genome and serine protease of a novel *Bacillus subtilis* isolate

Zahoor Khan · Maryam Shafique · Faizan Saleem · M. Kamran Azim ·
Nusrat Jabeen · Sehar Afshan Naz

Received: 9 February 2021 / Accepted: 30 November 2021 / Published online: 15 January 2022
© The Author(s), under exclusive licence to Springer Nature Switzerland AG 2021

Abstract Current study was undertaken to carry out the genome-wide analysis of a multipotent isolate from desert soil which was previously identified as *Bacillus tequilensis* based on 16S rDNA analysis. This study also aims to characterize the serine protease and its biocatalytic potentials implying a combination of empirical and in-silico approaches. Next generation sequencing and short read de novo assembly generated the 4,235,084 bp draft genome of *Bacillus* sp. ZMS-2. Genome sequence analysis by digital DNA:DNA hybridization (dDDH) and average nucleotide identity classified the isolate as *Bacillus subtilis* ZMS-2 (Bioproject ID: PRJNA691551). Genome annotation revealed 10 antibiotic resistance genes, 8 antibiotic/antifungal gene clusters and 25 genes encoding proteases including subtilisin E, an extracellular

alkaline protease. This extracellular protease (ZMS-2 protease) was produced using a statistically optimized medium, purified partially and characterized as alkaline serine protease. The partially purified ZMS-2 protease (780 U/mL) showed a 21 mm zone of casein hydrolysis and dehaired goat skin by pulling out hair with roots. These catalytic potentials of ZMS-2 protease were further confirmed using scanning electron microscopy of casein beads and dehaired skin. The study concludes *B. subtilis* ZMS-2 as a potent producer of a protease with promising potentials of commercial importance.

Keywords *Bacillus subtilis* ZMS-2 · Genome-wide · ZMS-2 protease · Caseinolytic · Keratinolytic

Z. Khan · N. Jabeen
Department of Microbiology, University of Karachi,
Karachi 75270, Pakistan

Z. Khan · M. Shafique (✉) · S. A. Naz
Department of Microbiology, Federal Urdu University of
Arts, Science and Technology, Karachi 75300, Pakistan
e-mail: maryamshafique@fuuast.edu.pk

F. Saleem · M. K. Azim
Department of Biosciences, Mohammad Ali Jinnah
University, Karachi 75400, Pakistan

F. Saleem
Department of Biology, McMaster University, Hamilton,
Ontario L8S 4L8, Canada

Introduction

Bacillus subtilis has been known as the most versatile soil bacterium with particular reference to its metabolic potential on the industrial scale (Earl et al. 2008). This bacterium is an obligate aerobe, however, can also survive anaerobically in the presence of glucose and nitrates owing to its ability for endospore formation to withstand extreme environmental conditions (Baysal and Yildiz 2017). Due to the feasibility of convenient exploitation of *B. subtilis* for genetic engineering as well as the potential to produce various bioactive molecules, it serves as an ideal candidate for

biotechnology and biopharmaceutical industry (Stein 2006). Besides these characteristic feature, *B. subtilis* has been given GRAS (Generally Recognized As Safe) status by the Food and Drug Administration of USA (FDA) due to the production of non-toxic by-products preferably the hydrolytic enzymes including proteases, lipases, cellulases, α -amylase, pullulanase and xylanases (Morikawa 2006). Moreover, it is a well-known producer of broad-spectrum antibiotics, fungicidal compounds and a potent bio-degrader of radioactive waste (Baysal and Yildiz 2017).

Proteolytic enzymes are naturally encoded by all organisms and are involved in a variety of physiological reactions ranging from food's proteins digestion to complex and regulated pathways like complement system, blood-clotting, apoptosis and the invertebrate Prophenoloxidase activating pathway (Bond 2019). *Bacillus* species, among all bacteria are extensively exploited microbes to produce alkaline proteases at commercial scale (Razzaq et al. 2019), which represent one of the three largest groups of commercial enzymes with a global enzyme market share of almost 60% (Ningthoujam et al. 2009). Serine proteases (EC 3.4.21) are endopeptidase characterized by the presence of serine residue in their catalytic triad (Gasteiger et al. 2003; Patel 2017). The mechanism of action of serine protease was discovered by studying the kinetics and catalysis of α -chymotrypsin in 1962 (Bender 1962). During catalysis the serine residue of protease forms an acyl-enzyme intermediate with substrate followed by its hydrolysis in the presence of water to release peptide chains (Rao et al. 1998). For a long time, only two clans of serine proteases were known; the subtilisin and the trypsin group. Advancements in the fields of molecular biology and bioinformatics led to the discovery of a wide variety of serine proteases (Polgár 2005) which are further categorized into several types on the basis of substrate specificity i.e., subtilisin, elastase, trypsin and chymotrypsin-like proteases along with few others as kallikrein and cathepsin (Gasteiger et al. 2003). Serine proteases usually remain active at neutral to alkaline pH and have low molecular mass of 18–35 kDa (Gupta et al. 2002). These enzymes perform momentous functions in various biological processes, like digestion, metabolism, signaling, apoptosis, immunity, blood coagulation, fertilization and regulation of development (Yang et al. 2015). During the last decade, microbial serine proteases have drawn interest of researchers'

due to their tremendous potential applications in leather processing, diagnostics and detergent formulations (Alemu 2015; Jisha et al. 2013). Current study is the continuation of our previous work where we characterized the isolate ZMS-2 as *Bacillus tequilensis* based on the phylogenetic analysis of 1,053 bases (Accession no. MK101013) of 16s rDNA gene (Khan et al. 2019). Moreover, the characterization of the whole genome was undertaken to further explore the potential commercial aspects and to reconfirm the phylogenetic affiliation of the producer strain. Here we report the whole genome draft, bio-catalytic potentials and recharacterization of *Bacillus tequilensis* ZMS-2 as *Bacillus subtilis* ZMS-2 using a combination of empirical and in-silico analyses.

Materials and methods

Materials

Skim milk (Skimmilac) from Millac Foods was purchased from local grocery shop, while the components of growth medium and azocasein/assay reagents used in this study were purchased from Oxoid Ltd. and Sigma-Aldrich, respectively. All other chemical/reagents used in this study were of analytical grade.

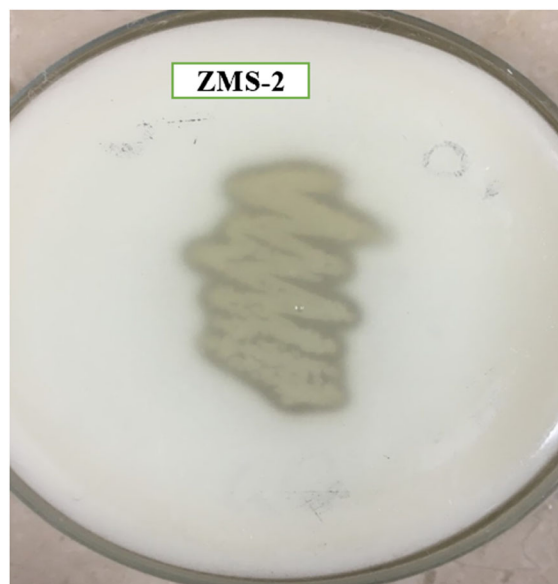


Fig. 1 a *Bacillus subtilis* ZMS-2 on skim milk agar, which was previously characterized as *Bacillus tequilensis* ZMS-2 (Khan et al. 2019)

Producer strain

Bacillus sp. ZMS-2 (Fig. 1, Genbank Accession No. MK101013) previously isolated from soil samples collected from hot desert of district Tharparkar, Sindh, Pakistan was subjected to genome wide analysis. Next Generation sequencing followed by genomic characterization was carried out to explore some new aspects of commercial significance and reconfirm its phylogenetic affiliation using 16S rDNA and MLST analysis. This strain is a potent producer of thermophilic alkaline protease having multifarious commercial catalytic potentials (Khan et al. 2019), which were further analyzed using scanning electron microscopy.

Genomic DNA purification and next generation sequencing

In order to extract the total genomic DNA, 2 mL of overnight culture was centrifuged at 10,000×g for 10 min and obtained cellular pellet was subjected to CTAB DNA extraction method (Nishiguchi et al. 2002). Qualitative and quantitative estimation of extracted DNA was performed by using agarose gel electrophoresis and QUBIT fluorometer (Invitrogen Inc., USA), respectively. Extracted DNA was then used for paired-end library construction by the use of Illumina TruSeq DNA PCR-Free kit by mechanical lysis of DNA followed by adapter ligation (insert size = 500 bp) and sequenced by using Illumina Hi-seq 2500 sequencing platform (Illumina Inc., San Diego, USA).

Genome assembly and annotation

Quality estimation of raw NGS sequences was performed by using FASTQC tool (Brown et al. 2017). Sequences having PHRED quality score less than 20 were removed by FASTX toolkit (http://hannonlab.cshl.edu/fastx_toolkit/). Quality filtered sequences were assembled into draft genome by using Unicycler prokaryotic genome assembler by generating assemble graphs on default range of k-mer sizes followed by selection of assembly having lower numbers of contigs with larger size and minimal dead end count (Wick et al. 2017) while quality of assembled contigs was analysed by QUAST tool (Gurevich et al. 2013). Genome annotation of assembled contigs was done by using RAST and PROKKA according to

default parameters (Aziz et al. 2008; Seemann 2014). COG orthology database was utilized for the annotation of coding sequences (Alexeyenko et al. 2006). Circular genome plot of annotated contigs was prepared by using CIRCOS tool by providing Genbank annotated file format along with associated contigs (Krzywinski et al. 2009). Psorb version 3.02 was used with default parameters for prediction of subcellular localization of proteases (Yu et al., 2010). Antibiotic resistance genes were identified using Resistance Gene Identifier (CARD database) with selection of perfect and strict hits only parameter and excluding Nudge ≥ 95% (Jia et al. 2016). Antibiotic/antifungal gene clusters were found by BAGEL4 server by defaults parameters of the server and three databases containing information related to classes I, II and II bacteriocins. (Van-Heel et al. 2018).

Phylogenetic analysis

The strain was previously identified as *Bacillus tequilensis* based on 16S rDNA analysis as the length of amplified sequence of 16S rDNA was only 1,053 bases (Accession no. MK101013). In order to reconfirm the identify genus level, contig coding for 16S rRNA gene (1550 bases) was aligned against NCBI 16S rRNA database by using BLAST sequence analysis tool (Madden 2013). For the identification on species level, Digital DNA:DNA hybridization (dDDH) and average nucleotide identity (ANI) was calculated by using type strain genotyping server (TYGS) and AutoMLST, respectively (Alanjary et al. 2019).

In-silico analysis of Subtilisin E

The biochemical properties and Signal peptide for the primary sequence of Subtilisin E were investigated using online ProtParam tool (Gasteiger et al. 2005) and SignalP 5.0 (Bendtsen et al. 2004), respectively.

Production of ZMS-2 protease

Physicochemical parameters (concentration of carbon and nitrogen sources, temperature, incubation time and agitation) for the maximum production of enzyme were optimized employing Plackett–Burman design followed by Box–Behnken design (data not shown). The designated strain was subjected to submerged

fermentation in 250 mL Erlenmeyer flasks using a statistically optimized production medium containing; 0.6% wheat bran, 0.4% casein, 0.5% starch, 0.1% peptone, 0.1% KH_2PO_4 and 0.1% MgSO_4 for 72 h at 56 rpm and 36.5 °C. After submerged fermentation, broth was centrifuged at 4000 rpm at 4 °C for 30 min and filtered using Whatman® filter paper to obtain cell free filtrate as source of extracellular protease. The proteolytic activity was assessed by colorimetric method using azocasein as substrate (Caldas et al. 2002). One unit of enzyme activity was defined as the amount of enzyme which yields an increase in absorbance of 0.001 at 440 nm in 30 min at 37 °C.

Partial purification and characterization of ZMS-2 protease

Cell free filtrate (100 mL) having 430 U/mL was subjected to gradual addition of ammonium sulfate up to 70% saturation level at 4 °C on magnetic stirrer. Precipitates recovered through centrifugation at 4000 rpm at 4 °C for 30 min and each pellet was dissolved in 200 μL of 50 mM Tris–HCl buffer (pH 8), dialyzed overnight (cut of 14 kDa) to remove access salts. The proteolytic units were estimated by colorimetric method using azocasein as substrate (Caldas et al. 2002). Proteolytic assay was carried out by adding 50 μL (1 and 5 mM) of each phenyl-methylsulfonyl fluoride (PMSF) and ethylene-diamine tetraacetic acid (EDTA), separately. The initial activity of ZMS-2 protease prior to the addition of inhibitors was taken as 100%.

Keratinolytic and caseinolytic activities of ZMS-2 protease

The keratinolytic potential of ZMS-2 protease was investigated by incubating a goat skin piece (2.5×2.0 cm) in 20 mL Tris-buffer (pH 8) containing 2 mL enzyme (780 U/mL) solution for 12 h in a 100 mL flask at 120 rpm and 37 °C. Similarly, as controls, two skin pieces were taken; one piece was shaved using Gillette blue III razor (during chemical dehairing the sulfide behaves like a blade), while the other one was incubated in the same buffer without addition of enzyme preparation. The caseinolytic activity of ZMS-2 protease was assessed using agar well diffusion method by adding 100 μL of enzyme (780 U/mL) to skim milk agar plate and incubated for

12–24 h at 37 °C. For control, 100 μL of Tris–HCl buffer was dispensed on the skim milk agar plate before incubation. Casein agar beads from zone of casein hydrolysis and zone having intact casein were taken using 5 mm sterilized borer and processed for microscopic analysis.

Topological analysis using scanning electron microscope (SEM)

Enzymatically dehaired goat skin, shaved skin, casein agar beads (hydrolyzed and non-hydrolyzed) were dried in an oven at 40–50 °C for 4–5 h. All samples were coated with 250 Å thick gold coating using Ion Sputtering Device (JEC-1500, Jeol, Japan) and analyzed by Analytical Scanning Electron Microscope (JSM-6380, Jeol, Japan). The images were taken at different magnifications ranging from 100 to 5000 $\times$ at a voltage of 10 kV.

Results

Genomic analysis of strain ZMS-2

The quality parameters for the assembled genome are mentioned in Table 1. Sequencing reads assembled into 30 contigs, with a total size of 4,235,084 bases. The N50 of the assembled genome was found to be 1,079,220 bp with the largest contig having a size of 1,137,525 bp. Figure 2 represents the draft genome map (4.2 Mbps) of strain ZMS-2. The whole genome data was submitted to NCBI having Bioproject ID: PRJNA691551.

Table 1 Genome assembly quality parameters of isolate ZMS2

Genome assembly parameters	Observations
No of contigs	30
Genome size (bp)	4,235,084
Contigs > 1000 bp	24
Largest contig	1,137,525
GC content (%)	43.30%
N50 (bp)	1,079,220
Number of N's per 100 kbp	0

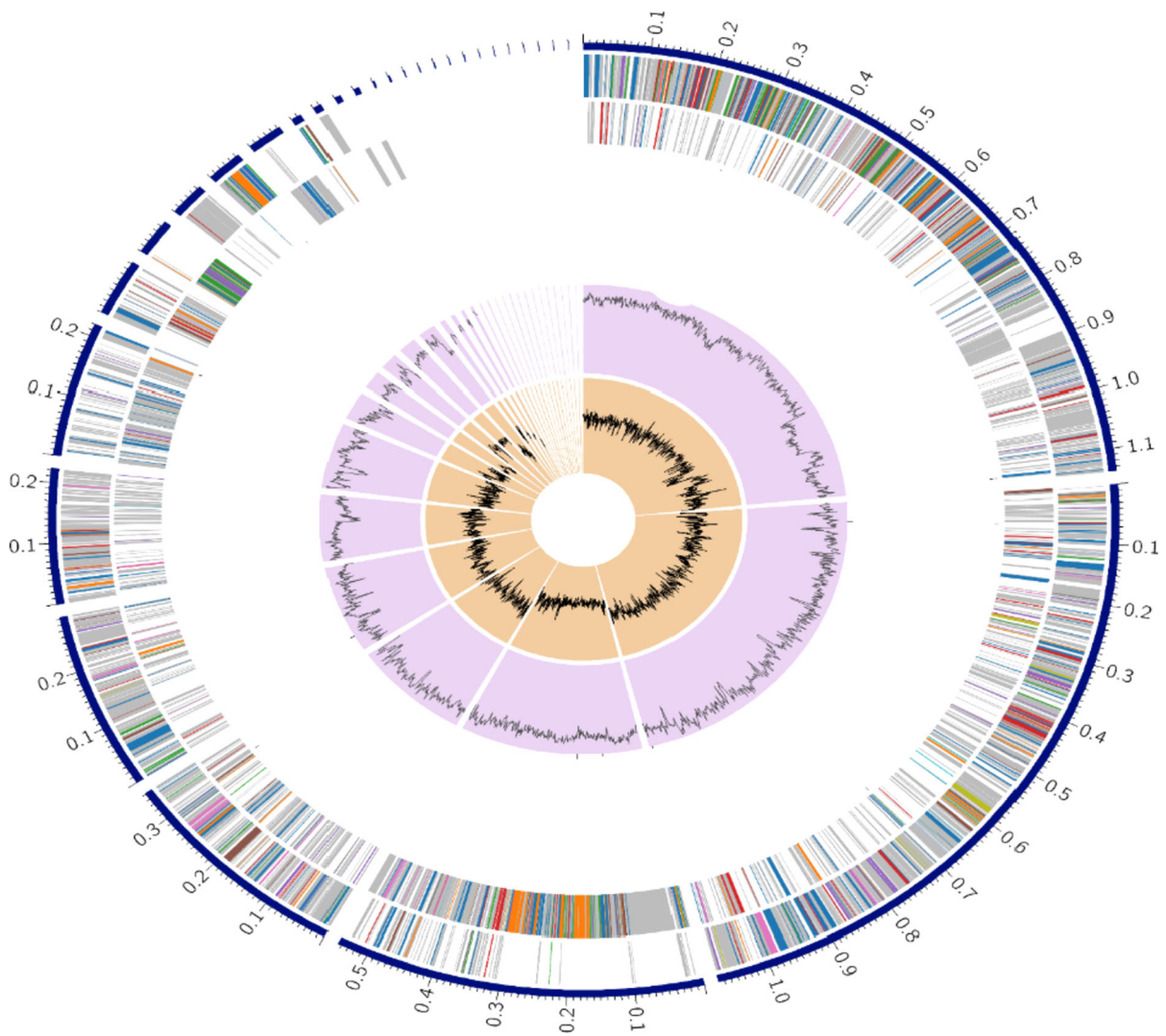


Fig. 2 CIRCOS plot (Krzywinski et al. 2009) of 30 annotated contigs of *Bacillus* sp. ZMS-2 isolate. The outermost circle represents the generated contigs followed by coding regions on forward-strand, coding regions on reverse-strand and GC skew

Phylogenetic characterization of strain ZMS-2

The contigs were aligned against NCBI 16S rRNA database and it was observed that contig 21 (1550 bases) possessed the 16S rRNA gene of the isolate ZMS-2. The sequence of 16S rRNA gene demonstrated the highest percentage identity (99.87%) against *Bacillus* species. This observation indicated the genus of strain ZMS-2 to be *Bacillus*. The strain ZMS-2 was characterized on species level using digital DNA:DNA hybridization (dDDH), Average nucleotide identity (ANI) and MLST based approaches. Figure 3 demonstrates the MLST based phylogenetic

tree for strain ZMS-2. It showed that strain ZMS-2 was closely associated with the branches of *Bacillus subtilis* species. The values for dDDH and ANI (Table 2) were found to be 92.9% and 98.38%, respectively.

Genome annotation

Functional annotation of the isolate ZMS-2 was performed by genome annotation pipelines (PROKKA and RAST). Total number of genes in the assembled genome was found to be 4449. Figure 4 represents the functional annotation of coding regions according to

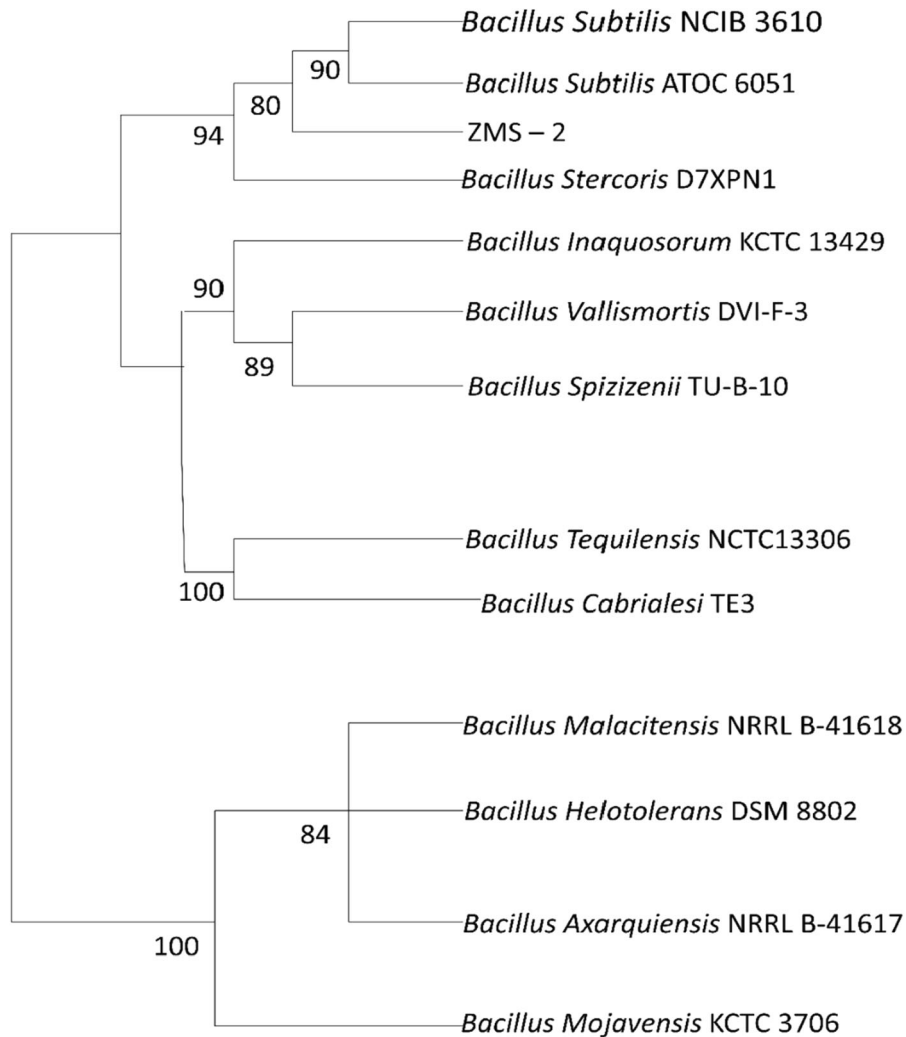


Fig. 3 MLST Based Phylogenetic Tree of 12 *Bacillus* species (including two *B. subtilis*) and *B. subtilis* ZMS-2

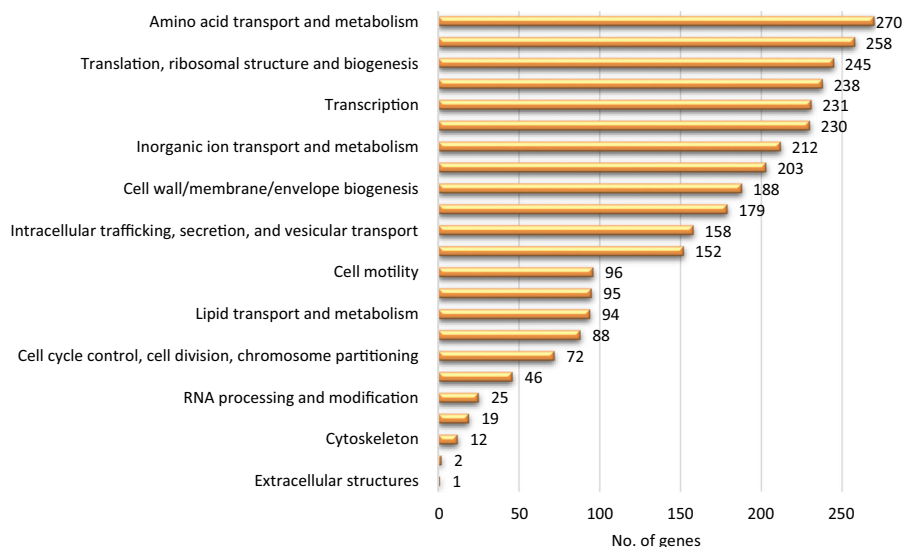
Table 2 Genome comparison *Bacillus subtilis* ZMS-2 and *Bacillus subtilis* ATCC-6051

Strain	Gene number	GC content (%)	tRNAs	rRNAs	ANI	Digi-DDH
<i>Bacillus subtilis</i> ZMS-2	4,449	43.3	60	3	98.38%	92.9%
<i>Bacillus subtilis</i> ATCC-6051	4391	43.6	71	5		

their metabolic association. Most genes were concomitant with amino acid transport and metabolism (9%), followed by energy production and conversion (8%), replication and repair (8%), post-translational modifications and protein turnover (7%) and carbohydrate metabolism (7%). Several genes correlated

with accessory metabolic pathways such as ion transport and metabolism (7%), secondary metabolite metabolism (3%), transport mechanisms (5%) etc.

Fig. 4 Annotation of *B. subtilis* ZMS-2 genes by the genome annotation pipelines (PROKKA and RAST) and their distribution according to different biochemical categories



Genomic characterization of genes encoding extracellular proteases

Genes coding for proteases in isolate ZMS-2 were annotated. Table 3 contains a list of 25 proteases identified in the genome sequence of the isolate ZMS-2 with predicted subcellular localization. Prediction of subcellular localization showed that 15 proteases were associated with cytoplasmic membrane, 7 proteases were of cytoplasmic region, while 3 proteases (i.e. Neutral protease B, Bacillolysin and Subtilisin E) were predicted as extracellular in nature.

Antibiotic/antifungal and antibiotic resistance gene clusters

Several antibiotic and antifungal gene clusters were also identified in isolate ZMS-2 (Table 4). Contig 2 possessed five antibiotic gene clusters (Subtilisin, Erisin S, Bacilysin, Cyclopeptide CDPS and Bacil-libactin), while contig 1 and contig 3 coded for bacteriocin and bacillaene, respectively. Contig 8 demonstrated the coding potential for antifungal agent (i.e. Fengycin). Table 5 contains the list of antibiotic resistance genes identified in isolate ZMS-2. Two genes (EbrA and EbrB) were found to code for multidrug resistance proteins, while two multidrug efflux pumps were associated with Lincomycin (LmrB) and Tetracycline (TetL) resistance. Additionally, resistance genes against Fosfomycin (FosB) and

Bacitracin (Bce gene cluster) were also detected in the genome of the isolate ZMS-2.

In-silico analysis of subtilisin E

Analysis of subtilisin E primary sequence by Prot-Param tool revealed that the ZMS-2 protease (Subtilisin E) was a polypeptide of 381 amino acids residues having an estimated molecular weight of 39.45 kDa. The assessed half-life was 30 h (mammalian reticulo-cytes, in vitro), > 20 h (yeast, in vivo) and > 10 h (*Escherichia coli*, in vivo). Furthermore, the instability index was computed to be 30.80, which suggests that the enzyme is stable. The aliphatic index, the Grand average of hydropathicity (GRAVY) and the isoelectric point for the enzyme were calculated as 81.23, − 0.062 and 9.04, respectively.

Signal peptide prediction

The Signal peptide for the ZMS-2 protease was predicted using SignalP 5.0, which revealed a cleavage site between residues at position no. 29 and 30 with a probability of 0.9369. Similarly, the likelihood scores for signal peptide (Sec/SPI), TAT signal peptide (Tat/SPI) and lipoprotein signal peptide (Sec/SPII) were 0.9929, 0.0025 and 0.0038, respectively.

Table 3 Genes encoding proteases and their predicted compartmentalization in isolate ZMS2

Gene Start	Gene end	Strand	Protease	Localization score	Subcellular localization
151,392	152,399	+	Protease IV	9.51	Cytoplasmic membrane
275,186	276,448	+	ATP-dependent Clp protease ATP-binding subunit ClpX	9.97	Cytoplasmic
296,057	296,803	+	Late competence protein ComC	10	Cytoplasmic membrane
366,000	363,925	–	Multimodular transpeptidase-transglycosylase	10	Cytoplasmic membrane
614,644	616,086	+	Carboxyl-terminal protease	9.81	Cytoplasmic membrane
692,207	691,614	–	ATP-dependent Clp protease proteolytic subunit ClpP	9.67	Cytoplasmic
872,441	873,817	+	Serine protease, DegP/HtrA, do-like	9.55	Cytoplasmic membrane
1,039,761	1,040,438	+	Putative membrane protease YugP	10	Cytoplasmic membrane
20,183	21,511	+	Serine protease AprX	9.6	Cytoplasmic
124,586	123,300	–	Uncharacterized zinc protease YmfH	7.5	Cytoplasmic
158,867	157,605	–	Intramembrane protease RasP/YluC	10	Cytoplasmic membrane
194,226	192,823	–	ATP-dependent hsl protease ATP-binding subunit HslU	9.97	Cytoplasmic
194,788	194,243	–	ATP-dependent protease subunit HslV	9.97	Cytoplasmic
341,306	342,871	+	Bacillolysins, extracellular neutral metalloprotease	9.98	Extracellular
443,373	445,472	+	ATP-dependent Clp protease, ATP-binding subunit ClpE	9.89	Cytoplasmic
466,109	465,213	–	Protease HtpX	10	Cytoplasmic membrane
494,119	495,078	+	Major intracellular serine protease precursor	10	Cytoplasmic
151,284	152,564	+	Uncharacterized metalloprotease YhfN	10	Cytoplasmic membrane
202,712	205,396	+	Cell wall-associated protease WprA	9.94	Cell wall
234,963	236,579	+	Extracellular neutral protease B	9.98	Extracellular
9644	8910	–	CAAX amino terminal protease family protein	10	Cytoplasmic membrane
49,405	50,022	+	Protease LexA	7.5	Cytoplasmic
57,233	54,801	–	ATP-dependent Clp protease, ATP-binding subunit ClpC	9.97	Cytoplasmic
16,291	17,163	+	Sporulation-specific protease YabG	7.5	Cytoplasmic
153,748	152,603	–	Subtilisin E	9.98	Extracellular

Production and partial purification of ZMS-2 protease

Crude protease extract (430 U/mL) was partially purified using ammonium sulfate precipitation technique and dialyzed overnight with 1.81-fold increase

in proteolytic activity (780 U/mL). Complete precipitation of enzyme was achieved at 70% ammonium sulfate precipitation. The enzyme was resuspended in 50 mM Tris–HCl buffer (pH 8) and further processed for characterization and application studies.

Table 4 Antibiotic and antifungal gene clusters characterized in genome of isolate ZMS2

Contig	Start	End	Percentage identity (%)	Class
1	1,041,620	1,061,824	100	Bacteriocin (UviB)
2	372,911	393,520	100	Subtilosin (SboX)
2	778,145	799,564	100	Erisin-S
2	323,424	364,842	100	Bacilysin
2	623,873	644,619	100	Cyclopeptide (CDPS)
2	947,622	997,363	100	Bacillibactin
3	13,890	119,134	100	Bacillaene
8	1	33,036	73	Fengycin

Table 5 Antibiotic resistance genes in genome of the isolate ZMS-2

Contig	Start	End	Gene symbol	Phenotype
1	424,879	425,733	ANT(6)-I	Aminoglycoside 6-nucleotidyltransferase
1	60,341	61,102	BceA	Bacitracin export ATP-binding protein
1	61,092	63,032	BceB	Bacitracin export permease protein
2	37,910	39,286	Tet(L)	Tetracycline resistance, MFS efflux pump
3	414,112	413,360	Vgb(B)	Streptogramin B lyase
3	17,515	17,835	EbrA	Multidrug resistance protein
3	17,849	18,202	EbrB	Multidrug resistance protein
7	112,443	113,876	Lmr(B)	Lincomycin resistance, MFS efflux pump
7	125,765	126,685	mphK	Macrolide 2'-Phosphotransferase I
8	50,994	50,560	FosB	Fosfomycin resistance protein

Biochemical properties of ZMS-2 protease

The ZMS-2 protease lost 29% of the proteolytic activity in presence of 50 μ L of 1 mM PMSF and lost 100% activity when 50 μ L of 5 mM PMSF (potent serine protease inhibitor) was added to the assay reagents. This inhibition of proteolytic activity in the presence of PMSF classify this enzyme as serine protease.

Keratinolytic and caseinolytic potentials of ZMS-2 protease

Partially purified ZMS-2 protease efficiently dehaired goat skin after 12 h of treatment in a shaker incubator at 37 °C. The loose hairs were plucked with fingers very easily along with root hair and epidermis was observed to be degraded efficiently. However, in control sample, the hairs were tightly bound, and it was even difficult to pull out hair with the help of forceps. The zone of hydrolysis of casein (21 mm) on skim milk agar plate was noted after 24 h of

incubation at 37 °C confirming the caseinolytic activity of ZMS-2 protease (Fig. 5).

Topological analysis of skin and agar beads

The surface morphology and texture of enzymatically dehaired and shaved goat skin pieces were analyzed using SEM (Fig. 6). Shaved skin displayed a rough surface, having intact hair roots and epidermis. On the contrary, enzyme assisted dehaired skin displayed a smooth surface with degraded epidermis and hair bulb having no hair. Surface topology of enzymatically dehaired skin confirmed efficient keratinolytic potential of ZMS-2 protease. The SEM images of agar beads from the zone of casein hydrolysis showed smooth homogenous morphology as compared to the intact beads of casein. Agar beads having intact casein displayed a rough surface even when observed at high resolution (Fig. 7c). Analysis of these microscopic images displayed effective caseinolytic potential of ZMS-2 protease (Fig. 7).

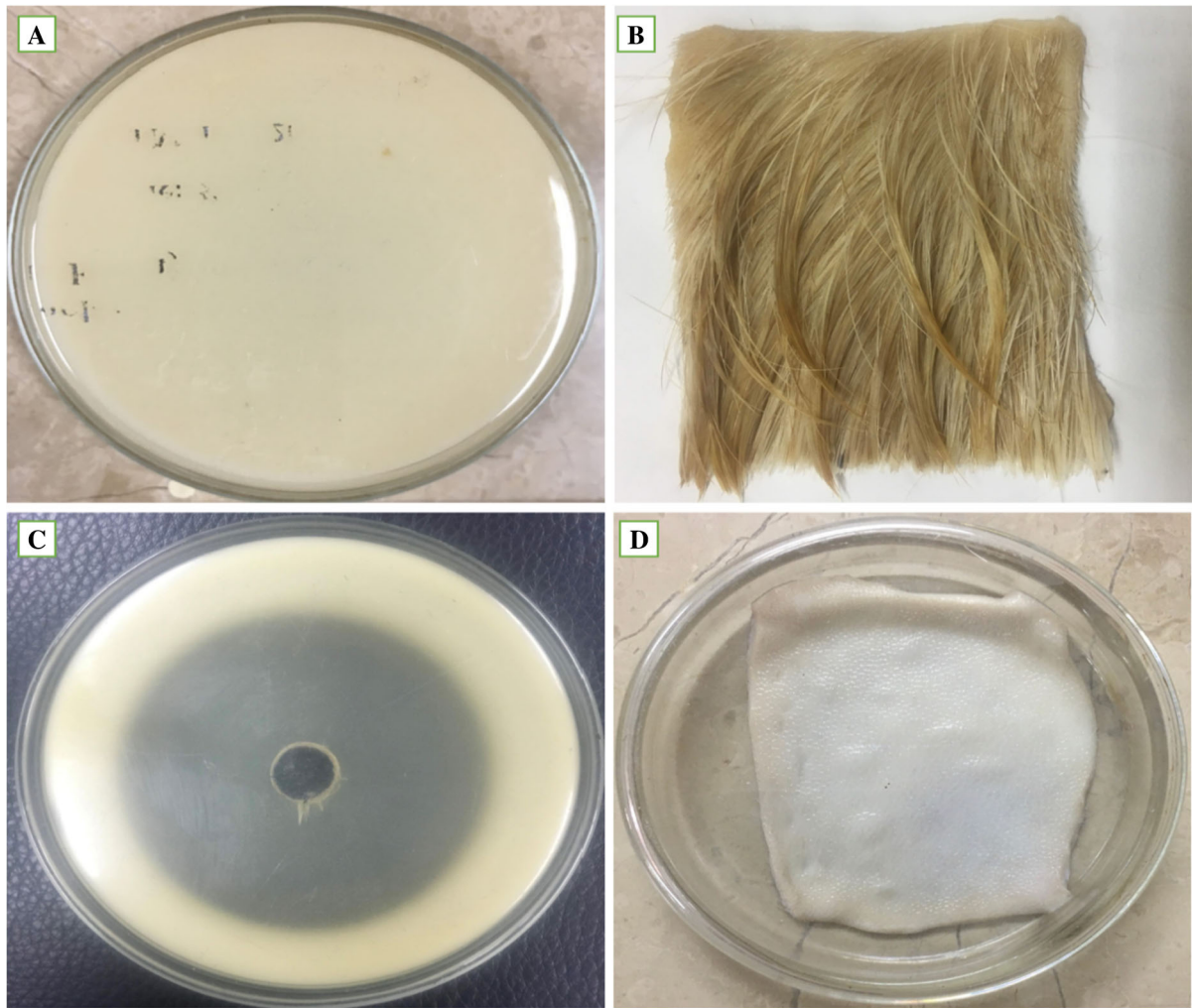


Fig. 5 Catalytic potential of ZMS-2 protease, A and C: Skim milk agar plate before and after casein hydrolysis by ZMS-2 protease. B and D: Goat skin before and after dehairing with ZMS-2 protease

Discussion

Current study highlighted the isolation, genome-wide sequencing and serine protease activity of a previously isolated strain from the indigenous desert soil samples. The genomic size of the isolate ZMS-2 was found as 4.2 Mbps (Fig. 2; Table 1). Digital DNA:DNA hybridization (DddH) value for isolate ZMS-2 against *Bacillus subtilis* was found to be 92.9%, while average nucleotide identity (ANI) value was 98.38%. Hence the isolate ZMS-2 was reidentified as a *Bacillus subtilis* as calculated dDDH value was > 70% and ANI value was > 94%. As it is noteworthy that the isolate ZMS-2 was previously identified as *Bacillus*

tequilensis (Khan et al. 2019) based on 16S rDNA phylogenetic analysis. However, the length of amplified sequence of 16S rDNA at that time was only 1,053 bases (Accession no. MK101013). During current study we have done the phylogenetic analysis based on 1550 bases of 16S rDNA for genus level as well as MLST analysis for the identification at species level.

Analysis of genome sequence showed 25 proteases/proteinases/peptidases divided into three groups based on subcellular localization (Table 3). Coding sequences for three extracellular proteases (Neutral protease B, Bacillolysin and Subtilisin E) were also found in genome of strain ZMS-2. Neutral protease B has been detected in several *Bacillus subtilis* species

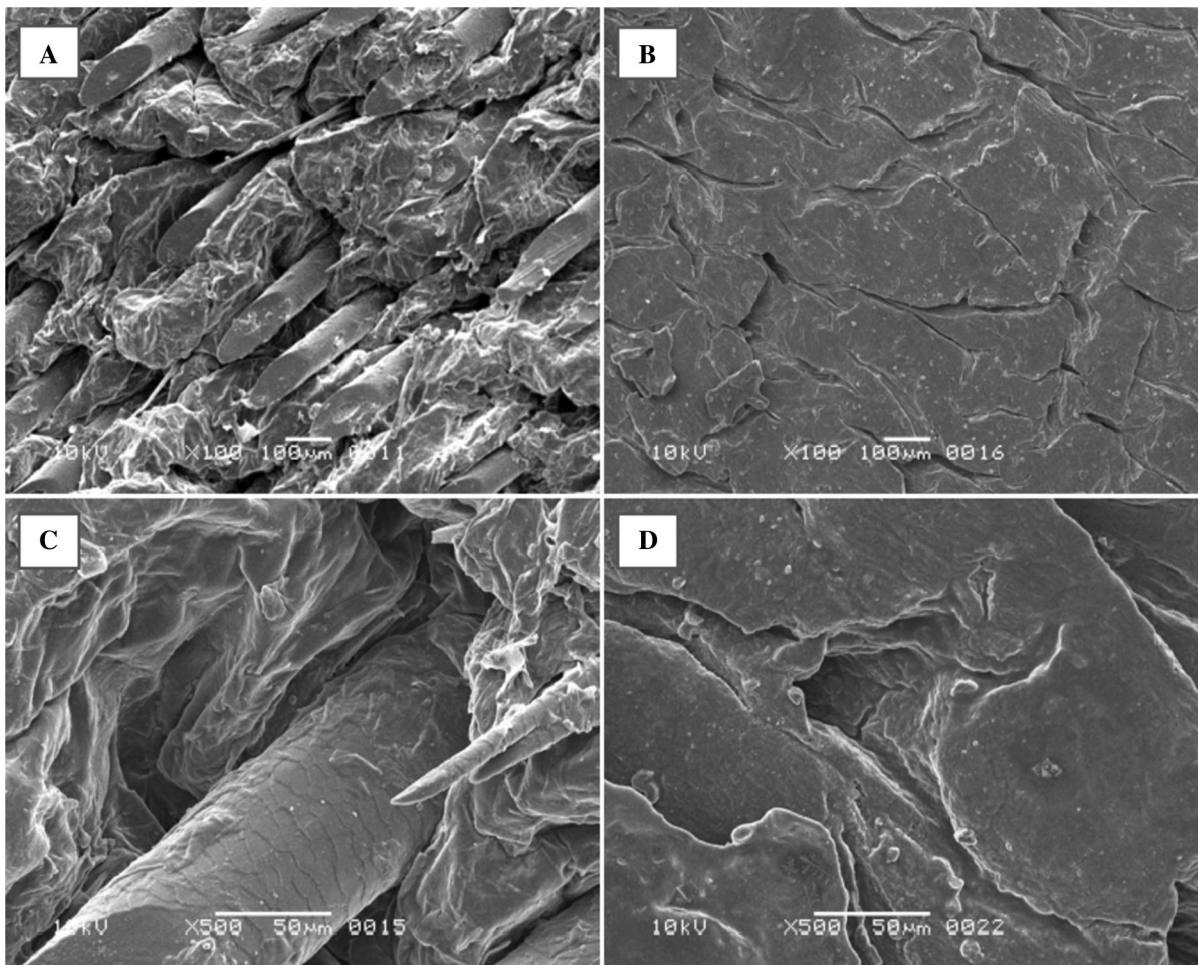


Fig. 6 Scanning electron microscopy of skin, **a** and **c**: SEM micrograph of shaved goat skin at 100× and 500× magnification respectively. **b** and **d**: SEM Micrograph of enzyme assisted dehaired skin at 100× and 500× magnification respectively

and demonstrates potent proteolytic activity at neutral pH (Cao et al. 2018). Bacillolysin has also been proven to possess goat skin dehairing potential and demonstrates highest activity in the presence of metallic ions (metalloprotease) and neutral pH (Tian et al. 2019).

The bacteriocin and antibiotics production potential of the isolate ZMS-2 can also be a promising aspect for its application as an industrial producer. More than two dozen of antibiotics and anti-microbial peptides are reported from *B. subtilis* including, lantibiotics, unusual lantibiotics (Subtilosin), and non-ribosomal biosynthesized peptides (bacillibactin). These antibiotics amazingly vary in their structures and an average of about 4–5% of a *B. subtilis* genome is devoted to antibiotic production. Furthermore, these antibiotics play distinct roles beyond the ‘pure’ anti-microbial

action: Non-ribosomally synthesized lipopeptides are involved in biofilm and swarming development, lantibiotics function as pheromones in quorum-sensing, and a ‘killing factor’ effectuates programmed cell death in sister cells (Stein 2006). The bacteriocins are antimicrobial peptides which have been a major area of scientific research due to their commercial importance as natural preservative and antibacterial agents (de la Fuente-Salcido et al. 2013).

The genome analysis of *B. subtilis* ZMS-2 revealed the coding potential for Fengycin, a biologically active cyclic lipopeptide having antifungal activity against filamentous fungi. It is synthesized as an immune response against fungal infection, and functions by damaging the target’s cell membrane. The safety profile and antifungal potential makes *B. subtilis* an

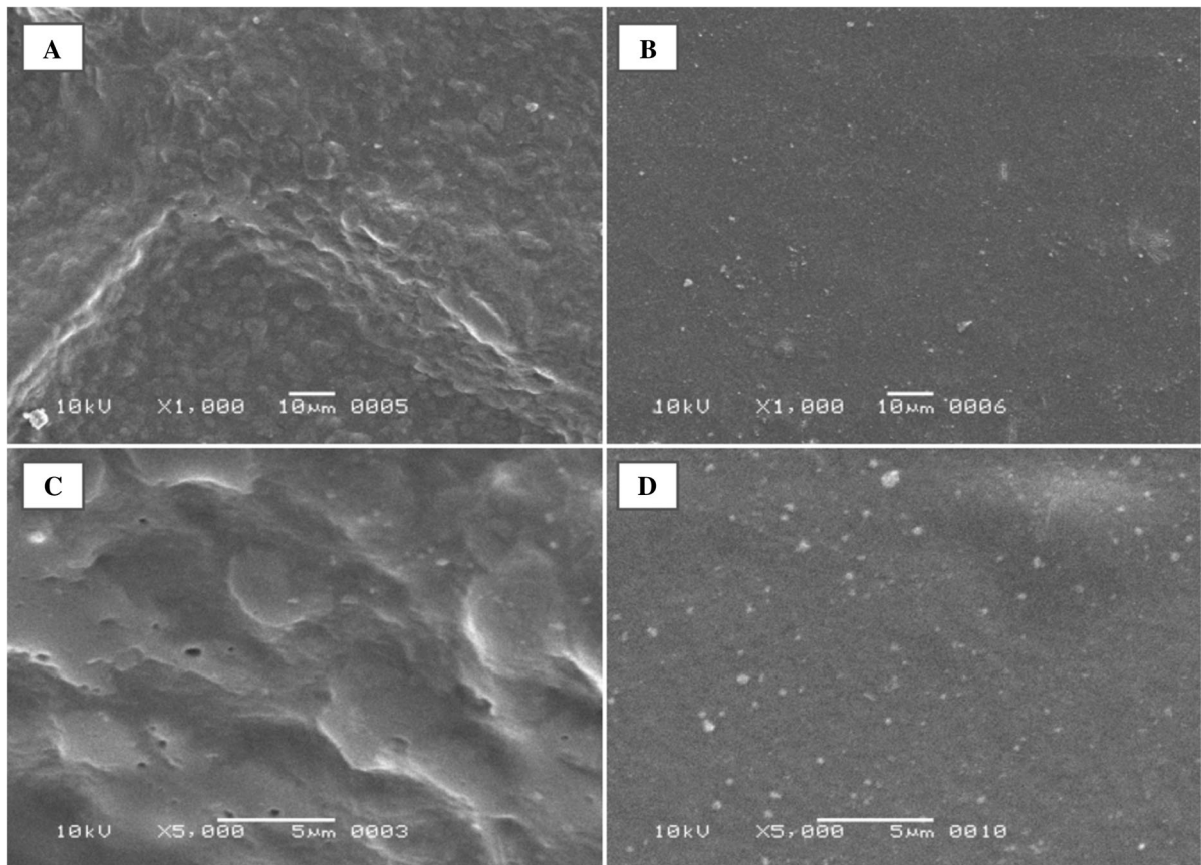


Fig. 7 Scanning electron microscopy of casein agar beads, **a** and **c**: SEM micrograph of casein intact agar bead at $\times 1000$ and $\times 5000$ magnification respectively. **b** and **d**: SEM

Micrograph of enzyme hydrolyzed casein agar bead at $\times 1000$ and $\times 5000$ magnification respectively

ideal natural fungicide and plant growth-promoting microorganism (Nicholson 2002).

ZMS-2 protease was partially purified using ammonium sulfate precipitation method which serves as an important step during protein purification ensuring bulk and quick precipitation of proteins (Burgess 2009). The enzyme was completely salted out at 70% saturation level with 1.81-fold purification. For various industrial applications, a partially purified enzyme can be a preferred choice due to its cost effectiveness.

The ZMS-2 protease showed optimum activity at alkaline pH, thermophilic (optimum activity at 60 °C) and extracellular in nature (Khan et al. 2019) and its proteolytic activity was completely inhibited by 50 μ L of 5 mM PMSF. The same properties were noted during the in-silico analysis of primary sequence of subtilisin E, suggesting that a component of the ZMS-2 protease is the extracellular subtilisin E.

Partially purified ZMS-2 protease preparation efficiently dehaired goat skin thereby confirming its keratinolytic potential which is in coherence with few previous studies (Shaikh et al. 2019; Huang et al. 2003; Zambare et al. 2007; Thanikaivelan et al. 2004; Sivasubramanian et al. 2008). It is important to note that not all proteases are applicable for dehairing applications due to their collagenolytic activity (Huang et al. 2003). Therefore, those proteases possessing efficient keratinolytic and minimal collagenolytic potential are ideal for tanneries (Zambare et al. 2007). During the process of dehairing, proteases selectively degrade the soft keratin tissue in the follicle, thereby pulling out intact hairs without disturbing tensile strength of the processed leather (Thanikaivelan et al. 2004). The excellent keratinolytic activity and no damage to the collagen structure as evident from the SEM analysis makes

ZMS-2 protease an ideal candidate for dehairing of animal's skins. Application of proteases in tanneries warranty partial or complete elimination of lime and sodium sulfide, thus helping to minimize the problem of stream pollution in addition to improved leather quality (Sivasubramanian et al. 2008).

Partially purified ZMS-2 protease displayed a 21 mm zone of casein hydrolysis on skim milk agar plate, confirming its caseinolytic potential. During casein hydrolysis, protease cleaves the peptide bonds within the polypeptide chains of complexed structure of casein, leading to generation of peptides of varying lengths. The caseinolytic potential of protease has immense commercial importance as casein hydrolysates are used as calcium binding peptides (Miao et al. 2016) and as stabilizers in the food industry (Alongi et al. 2014).

Scanning electron microscopic images displayed a major difference in the topology of shaved and enzyme assisted dehaired skin. Enzymatically dehaired skin when observed at high resolution (500×) displayed degraded epidermis and degraded hair bulb which is essentially important for the pulling out of hair along with the root (Fig. 6d). Furthermore, this degradation could possibly cut down the succeeding step of bating during leather processing as dehairing and bating was simultaneously done with ZMS-2 protease as a single step. Analysis of SEM images confirmed efficient keratinolytic potential of ZMS-2 protease. SEM micrograph at higher resolution (5000×) of casein intact beads displayed a rough and heterogeneous surface which might be due to the presence of intact casein molecules (Fig. 7c). The presence of globular shaped particles was observed at the surface of casein hydrolyzed beads at higher resolution (5000×) indicating the presence of ZMS-2 protease (Fig. 7d). This caseinolytic potential of ZMS-2 protease can be rightly exploited for applications in food, pharmaceuticals and chemical industries.

Conclusion

Current study highlights the recharacterization of isolate ZMS-2 as *Bacillus subtilis*, a strain with dynamic potential to produce extracellular serine protease as well as various other bioactive metabolites. This extracellular serine protease produced by ZMS-2 strain when partially purified displayed excellent

keratinolytic and caseinolytic activities. These findings suggest the scale up production and pilot scale application of this enzyme for the replacement of hazardous chemicals like sulfide and lime conventionally used in leather processing. This protease can also serve as a powerful tool for green processing in food, feed and biochemical industries.

Acknowledgements The authors acknowledge Higher Education Commission, Pakistan for the provision of funds for this study through Technology Development Fund (TDF) Grant No. 02-078 (awarded to corresponding author) and for Indigenous Ph.D. fellowship (awarded to first author).

Authors' contributions ZK Sampling, Experimental work, application of protease and manuscript writing. Dr. MS Experimental design, data analysis, interpretation of results and manuscript writing. FS data analysis and manuscript writing. MKA interpretation of results, data analysis and critical review. Dr. NJ Critical review and data analysis. Dr. SAN Critical review and data analysis.

Funding The authors acknowledge Higher Education Commission, Pakistan for the provision of funds for this study through TDF Grant No. 02-078 (awarded to corresponding author) and for Indigenous Ph.D. fellowship (awarded to first author).

Availability of data and material Whole genome data is submitted to NCBI having Bioproject ID: PRJNA691551 and is freely available. Moreover, all relevant raw data, will be freely available to any researcher wishing to use it for non-commercial purposes, without breaching participant confidentiality.

Code availability Not applicable.

Declarations

Conflict of interest The authors declare no conflict of interests.

Ethical approval Not applicable.

Consent to participate Not applicable.

Consent for publication Not applicable.

References

- Alanjary M, Steinke K, Ziemert N (2019) AutoMLST: an automated web server for generating multi-locus species trees highlighting natural product potential. *Nucl Acids Res* 47(W1):276–282
- Alemu F (2015) Isolation and screening of protease enzyme producing bacteria from cheese at Dilla University, Ethiopia. *Int J Food Sci Nutr* 4:234–239

- Alexeyenko A, Lindberg J, Pérez-Bercoff Á, Sonnhammer EL (2006) Overview and comparison of ortholog databases. *Drug Discov Today Technol* 3:137–143
- Alongi J, Carletto RA, Bosco F et al (2014) Caseins and hydrophobins as novel green flameretardants for cotton fabrics. *Polym Degrad Stab* 99:111–117
- Aziz RK, Bartels D, Best AA et al (2008) The RAST server: rapid annotations using subsystems technology. *BMC Genomics* 9:75
- Baysal O, Yildiz A (2017) *Bacillus Subtilis*: an industrially important microbe for enzymes production. *EC Microbiol* 5(4):148–156
- Bender ML (1962) The mechanism of α -chymotrypsin-catalyzed hydrolyses 1–3. *J Am Chem Soc* 84(13):2582–2590
- Bendtsen JD, Nielsen H, von Heijne G, Brunak S (2004) Improved prediction of signal peptides: SignalP 3.0. *J Mol Biol* 340(4):783–795
- Bond JS (2019) Proteases: history, discovery and roles in health and disease. *J Biol Chem* 294(5):1643–1651
- Brown J, Pirrung M, McCue LA (2017) FQC Dashboard: integrates FastQC results into a web-based, interactive, and extensible FASTQ quality control tool. *Bioinformatics* 33:3137–3139
- Burgess RR (2009) Protein precipitation techniques. *Methods Enzymol* 463:331–342
- Caldas C, Cherqui A, Pereira A, Simoes N (2002) Purification and characterization of an extracellular protease from *Xenorhabdus nematophila* involved in insect immune suppression. *Appl Environ Microbiol* 68:1297–1304
- Cao H, Liu L, Xu F, Yu J, Ye T, Yuan M (2018) Immobilization of neutral protease from *Bacillus Subtilis* via a high-affinity ligand. *Chem Lett* 47(4):464–467
- Earl AM, Losick R, Kolter R (2008) Ecology and genomics of *Bacillus subtilis*. *Trends Microbiol* 16:269–275
- de la Fuente-Salcido NM, Casados-Vázquez LE, Barboza-Corona JE (2013) Bacteriocins of *Bacillus thuringiensis* can expand the potential of this bacterium to other areas rather than limit its use only as microbial insecticide. *Can J Microbiol* 59:515–522
- Gasteiger E, Gattiker A, Hoogland C, Ivanyi I, Appel RD, Bairoch A (2003) ExPASy: the proteomics server for in-depth protein knowledge and analysis. *Nucl Acids Res* 31(13):3784–3788
- Gasteiger E, Hoogland C, Gattiker A et al (2005) Protein identification and analysis tools on the expasy server, 11th edn. Hum Publication, Totowa, pp 571–607
- Gupta R, Beg Q, Lorenz P (2002) Bacterial alkaline proteases: molecular approaches and industrial applications. *Appl Microbiol Biotechnol* 59(1):15–32
- Gurevich A, Saveliev V, Vyahhi N, Tesler G (2013) QUASt: quality assessment tool for genome assemblies. *Bioinformatics* 29(8):1072–1075
- Huang Q, Peng Y, Li X (2003) Purification and characterization of extracellular alkaline serine protease with dehairing function from *Bacillus pumilis*. *Curr Microbiol* 43:169–173
- Jia B, Raphenya AR, Alcock B et al (2016) CARD 2017: expansion and model-centric curation of the comprehensive antibiotic resistance database. *Nucl Acids Res* gkw1004
- Jisha VN, Smitha RB, Pradeep S, Sreedevi S, Unni KN, Sajith S, Priji P, Josh MS, SailasBenjaminet S (2013) Versatility of microbial proteases. *Adv Enzyme Res* 1:39–51
- Khan Z, Shafique M, Nawaz HR, Jabeen N, Naz SA (2019) *Bacillus tequilensis* ZMS-2: a novel source of alkaline protease with antimicrobial, anti-coagulant, fibrinolytic and dehairing potentials. *Pak J Pharm Sci* 32(4):1913–1918
- Krzywinski M, Schein J, Birol I et al (2009) Circos: an information aesthetic for comparative genomics. *Genome Res* 19(9):1639–1645
- Madden T (2013) The BLAST sequence analysis tool. In: The NCBI handbook [Internet], 2nd edn. National Center for Biotechnology Information (US)
- Miao J, Liu G, Luo Z et al (2016) Calcium-binding peptides isolated from casein hydrolysates enhances intestinal calcium uptake. *FASEB J* 30(690):1015
- Morikawa M (2006) Beneficial biofilm formation by industrial bacteria *Bacillus subtilis* and related species. *J Biosci Bioeng* 101(1):1–8
- Nicholson WL (2002) Roles of *Bacillus* endospores in the environment. *Cell Mol Life Sci* 59:410–416
- Ningthoujam DS, Kshetri P, Sanasam S, Nimaichand S (2009) Screening, identification of best producers and optimization of extracellular proteases from moderately halophilic alkalithermotolerant indigenous actinomycetes. *World Appl Sci J* 7:907–916
- Nishiguchi MK, Doukakis P, Egan M et al (2002) DNA isolation procedures. In: DeSalle R, Giribet G, Wheeler W (eds) *Techniques in molecular systematics and evolution*. Birkhäuser, Basel, pp 249–287
- Patel S (2017) A critical review on serine protease: key immune manipulator and pathology mediator. *Allergol Immunopathol* 45(6):579–591
- Polgár L (2005) The catalytic triad of serine peptidases. *Cell Mol Life Sci* 62(19–20):2161–2172
- Rao MB, Tanksale AM, Ghatge MS, Deshpande VV (1998) Molecular and biotechnological aspects of microbial proteases. *Microbiol Mol Biol Rev* 62:597–635
- Razzaq A, Shamsi S, Ali A, Ali Q, Sajjad M, Malik A, Ashraf M (2019) Microbial proteases applications. *Front Bioeng Biotechnol* 7:110
- Seemann T (2014) Prokka: rapid prokaryotic genome annotation. *Bioinformatics* 30:2068–2069
- Shaikh M, Shafique M, Naz SA et al (2019) *Streptomyces* sp. MM-3 from rhizosphere of Psidium Guajava: a potential candidate for protease with dehairing properties. *Pak J Bot* 51(2):735–742
- Sivasubramanian S, Murali Manohar B, Rajaram A, Puvanakrishna R (2008) Ecofriendly lime and sulfide free enzymatic dehairing of skins and hides using a bacterial alkaline protease. *Chemosphere* 70(6):1015–1024
- Stein T (2006) *Bacillus subtilis* antibiotics: structures, syntheses and specific functions. *Mol Microbiol* 56(4):845–857
- Thanikaivelan P, Rao JR, Nair BU, Ramasami T (2004) Progress and recent trends in biotechnological methods for leather processing. *Trends Biotechnol* 22(4):181–188
- Tian J, Long X, Tian Y, Shi B (2019) Eco-friendly enzymatic dehairing of goatskins utilizing a metalloprotease highly expressed by *Bacillus subtilis* SCK6. *J Clean Prod* 212:647–654

- van Heel AJ, de Jong A, Song C, Viel JH, Kok J, Kuipers OP (2018) BAGEL4: a user-friendly web server to thoroughly mine RiPPs and bacteriocins. *Nucl Acids Res* 46(W1):278–281
- Wick RR, Judd LM, Gorrie CL, Holt KE (2017) Unicycler: resolving bacterial genome assemblies from short and long sequencing reads. *PLoS Comput Biol* 13(6):e1005595
- Yang Y, Wen JY, Cai YN, Vallée I, Boireau P, Liu MY, Cheng SP (2015) Serine proteases of parasitic helminths. *Korean J Parasitol* 53(1):1–11
- Yu NY, Wagner JR, Laird MR, Melli G, Rey S, Lo R, Brinkman FS (2010) PSORTb 3.0: improved protein subcellular localization prediction with refined localization subcategories and predictive capabilities for all prokaryotes. *Bioinformatics* 26(13):1608–1615
- Zambare VP, Nilegaonkar SS, Kanekar PP (2007) Production of an alkaline protease by *Bacillus cereus* MCM B-326 and its application as a dehairing agent. *World J Microbiol Biotechnol* 23:1569–1574
- Publisher's Note** Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.